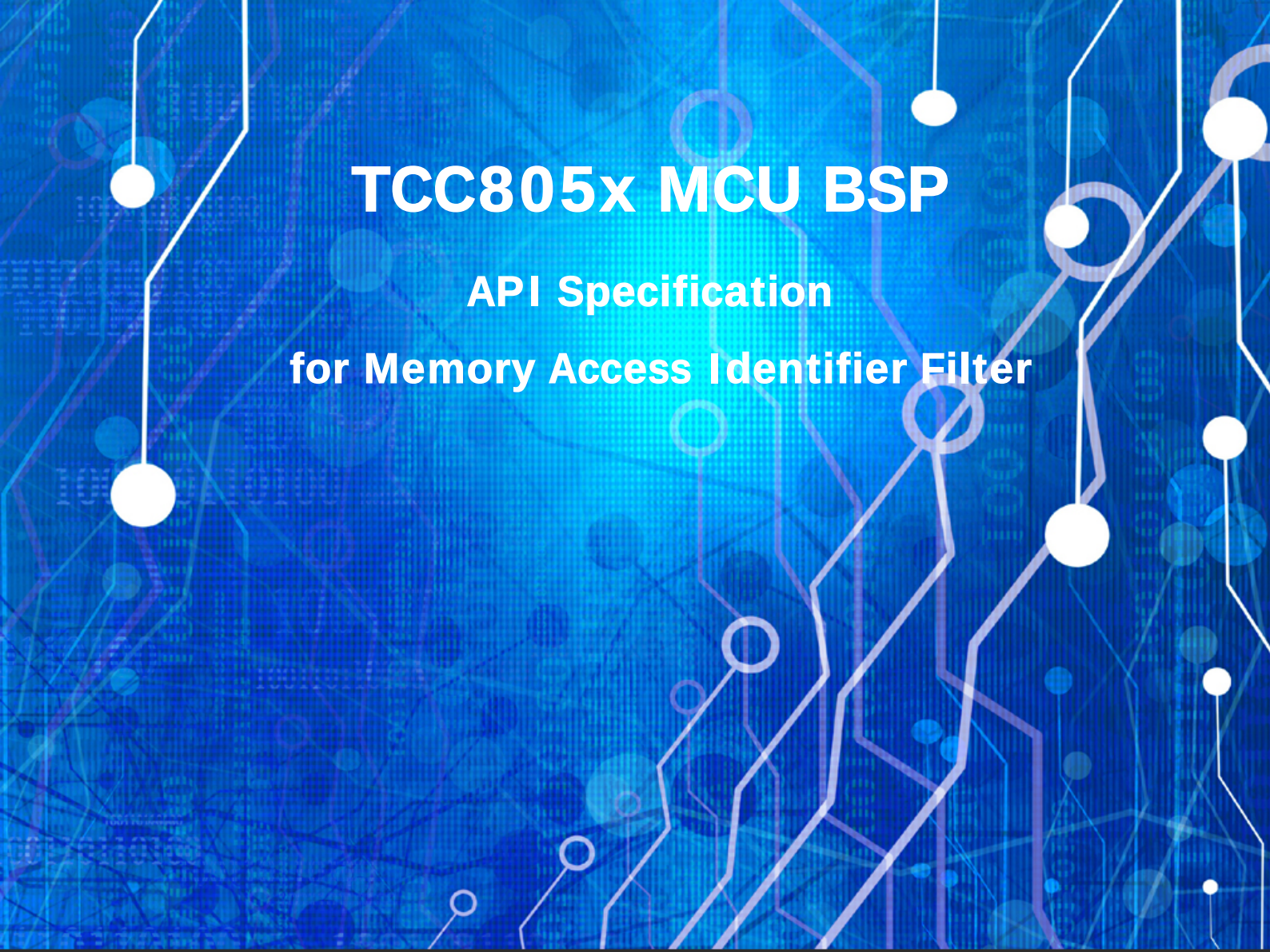




TCC805x MCU BSP

API Specification

for Memory Access Identifier Filter

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the memory access identifier filter device driver (hereinafter referred to as MIDF driver) of TCC805x MCU BSP. For more detailed information of the memory access identifier filter in the TCC805x MCU Sub-system, refer to Chapter 8 Memory Access ID (MID) Filter of the *“PART 11-MICOM SUB-SYSTEM”* in the *“TCC805x Full Specification”*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

MIDF	Memory Access Identifier Filter
-------------	---------------------------------

3 SOURCE FILES

3.1 Location of source Files

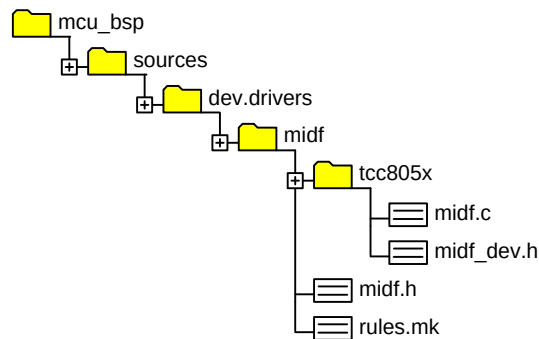


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- midf.c
 - MIDF driver source
- midf_dev.c
 - MIDF hardware definition header
- midf.h
 - MIDF driver API header
- rules.mk
 - Make rules

4 CONSTANT

4.1 Memory Access ID

Constant variables for identifying each bus master.

The MIDF hardware (TZC-400 of ARM) supports only 16 IDs, but TCC805x SoC has total 20 bus masters. Therefore, the same ID is allocated for GPSB3, 4, 5 and UART3, 4, 5, by default. If you want to distinguish each GPSB 3, 4, 5 channel, you can modify the following definition—deleting a definition of unused bus master.

Declaration

```
#define MIDF_ID_DMA0      (0UL)
#define MIDF_ID_DMA1      (1UL)
#define MIDF_ID_CAN0      (2UL)
#define MIDF_ID_CAN1      (3UL)
#define MIDF_ID_CAN2      (4UL)
#define MIDF_ID_GPSB0      (5UL)
#define MIDF_ID_GPSB1      (6UL)
#define MIDF_ID_GPSB2      (7UL)
#define MIDF_ID_GPSB3_4_5 (8UL)
#define MIDF_ID_UART0      (9UL)
#define MIDF_ID_UART1     (10UL)
#define MIDF_ID_UART2     (11UL)
#define MIDF_ID_UART3_4_5 (12UL)
#define MIDF_ID_CR5        (13UL)
#define MIDF_ID_JTAG       (14UL)
#define MIDF_ID_AP         (15UL)
```

Description

Value	Description
<i>MIDF_ID_DMA0</i>	<i>Memory access ID for DMA channel 0</i>
<i>MIDF_ID_DMA1</i>	<i>Memory access ID for DMA channel 1</i>
<i>MIDF_ID_CAN0</i>	<i>Memory access ID for CAN channel 0</i>
<i>MIDF_ID_CAN1</i>	<i>Memory access ID for CAN channel 1</i>
<i>MIDF_ID_CAN2</i>	<i>Memory access ID for CAN channel 2</i>
<i>MIDF_ID_GPSB0</i>	<i>Memory access ID for GBPS channel 0</i>
<i>MIDF_ID_GPSB1</i>	<i>Memory access ID for GBPS channel 1</i>
<i>MIDF_ID_GPSB2</i>	<i>Memory access ID for GBPS channel 2</i>
<i>MIDF_ID_GPSB3_4_5</i>	<i>Memory access ID for GBPS channel 3, 4, 5</i>
<i>MIDF_ID_UART0</i>	<i>Memory access ID for UART channel 0</i>
<i>MIDF_ID_UART1</i>	<i>Memory access ID for UART channel 1</i>
<i>MIDF_ID_UART2</i>	<i>Memory access ID for UART channel 2</i>
<i>MIDF_ID_UART3_4_5</i>	<i>Memory access ID for UART channel 3, 4, 5</i>
<i>MIDF_ID_CR5</i>	<i>Memory access ID for Cortex-R5 CPU</i>
<i>MIDF_ID_JTAG</i>	<i>Memory access ID for J-Tag</i>
<i>MIDF_ID_AP</i>	<i>Memory access ID for AP(A72, A53, M4) bus</i>

Remark

Do not change the definition of MIDF_ID_CF5, MIDF_ID_JTAG, and MIDF_ID_AP.

4.2 Memory Access Identifier Region Minimum Size

Constants values for the minimum region size can be protected by MIDF. The MIDF only permits a region to be started at 4KB-aligned address.

Declaration

```
#define MIDF_REGION_MIN_SIZE (0x1000UL)
```

Description

<i>Value</i>	<i>Description</i>
<i>MIDF_REGION_MIN_SIZE</i>	<i>Minimum length of the region</i>

Remark

None

4.3 MIDF Action Type

Definition of memory accessing attribute for reading and writing.

Declaration

```
#define MIDF_TYPE_READ (1UL << 0)
#define MIDF_TYPE_WRITE (1UL << 1)
```

Description

<i>Value</i>	<i>Description</i>
<i>MIDF_TYPE_READ</i>	<i>Memory accessing attribute for reading</i>
<i>MIDF_TYPE_WRITE</i>	<i>Memory accessing attribute for writing</i>

Remark

None

4.4 MIDF Return Type

Definition of MIDF driver's return value.

Declaration

```
#define MIDF_SUCCESS          (0L)
#define MIDF_ERROR            (-1L)
#define MIDF_ERROR_NOT_INITIALIZE (-2L)
#define MIDF_ERROR_NO_RESOURCE (-3L)
#define MIDF_ERROR_BAD_PARAMETERS (-4L)
#define MIDF_ERROR_BAD_RANGES (-5L)
#define MIDF_ERROR_NOT_FOUND  (-6L)
```

Description

Value	Description
<i>MIDF_SUCCESS</i>	<i>Operation success</i>
<i>MIDF_ERROR</i>	<i>Operation failure</i>
<i>MIDF_ERROR_NOT_INITIALIZE</i>	<i>Not initialized</i>
<i>MIDF_ERROR_NO_RESOURCE</i>	<i>Region already assigned</i>
<i>MIDF_ERROR_BAD_PARAMETERS</i>	<i>Bad parameter</i>
<i>MIDF_ERROR_BAD_RANGES</i>	<i>Memory range overlapped</i>
<i>MIDF_ERROR_NOT_FOUND</i>	<i>Cannot find the region</i>

Remark

None

5 APIs

5.1 MIDF_FilterInitialize

Initialize the MIDF. This API configures the memory ID for each bus master and sets the region zero(0) to be accessible to Cortex-R5 and J-TAG bus for whole memory region(0x00000000 ~ 0xFFFFFFFF).

Declaration

```
void MIDF_FilterInitialize  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

5.2 MIDF_RegisterFilterRegion

Register the MIDF filter region with a given parameter.

Declaration

```
int32 MIDF_RegisterFilterRegion
(
    uint32 uiMemoryBase,
    uint32 uiSize,
    uint32 uiID,
    uint32 uiType
);
```

Parameters

Value	Description
<i>uiMemoryBase</i>	<i>[in] Base address of memory buffer for 'id'. It should be 4 KB aligned.</i>
<i>uiSize</i>	<i>[in] Size of memory buffer for 'id'. It should be 4 KB aligned.</i>
<i>uiID</i>	<i>[in] Master id</i>
<i>uiType</i>	<i>[in] Read or Write type of memory buffer for 'id'</i>

Return

Value	Description
<i>MIDF_SUCCESS</i>	<i>Operation success</i>
<i>MIDF_ERROR</i>	<i>Operation failure</i>
<i>MIDF_ERROR_NOT_INITIALIZE</i>	<i>Not initialized</i>
<i>MIDF_ERROR_NO_RESOURCE</i>	<i>Region already assigned</i>
<i>MIDF_ERROR_BAD_PARAMETERS</i>	<i>Bad parameter</i>
<i>MIDF_ERROR_BAD_RANGES</i>	<i>Memory range overlapped</i>
<i>MIDF_ERROR_NOT_FOUND</i>	<i>Cannot find the region</i>

Remark

None

5.3 MIDF_UnregisterFilterRegion

Unregister the MIDF filter region with a given parameter. The parameters require the same value with the registered one to search and confirm the region.

Declaration

```
int32 MIDF_UnregisterFilterRegion
(
    uint32 uiMemoryBase,
    uint32 uiSize,
    uint32 uiID,
    uint32 uiType
);
```

Parameters

Value	Description
<i>uiMemoryBase</i>	<i>[in] Base address of memory buffer for 'id'. It should be 4kb aligned.</i>
<i>uiSize</i>	<i>[in] Size of memory buffer for 'id'. It should be 4kb aligned.</i>
<i>uiID</i>	<i>[in] Master id</i>
<i>uiType</i>	<i>[in] Read or Write type of memory buffer for 'id'</i>

Return

Value	Description
<i>MIDF_SUCCESS</i>	<i>Operation success</i>
<i>MIDF_ERROR</i>	<i>Operation failure</i>
<i>MIDF_ERROR_NOT_INITIALIZE</i>	<i>Not initialized</i>
<i>MIDF_ERROR_NO_RESOURCE</i>	<i>Region already assigned</i>
<i>MIDF_ERROR_BAD_PARAMETERS</i>	<i>Bad parameter</i>
<i>MIDF_ERROR_BAD_RANGES</i>	<i>Memory range overlapped</i>
<i>MIDF_ERROR_NOT_FOUND</i>	<i>Cannot find the region</i>

Remark

None

6 HOW TO USE MIDF DRIVER

In order to use the MIDF driver, the ***MIDF_FilterInitialize()*** function must be called first. You can set up the memory filtering function for a specific memory area by using ***MIDF_RegisterFilterRegion()*** function. In 6.1 sample code, it is assumed that CAN0 bus needs to access only 4Kbytes length of memory region starting from 0xC100000.

6.1 Initialize and Register /Unregister Region Filter

```
Int main()
{
    uint32 uiMemoryBase   = 0xC1000000; //SRAM1
    uint32 uiSize         = 0x1000; //4KB
    uint32 uiID           = MIDF_ID_CAN0;
    uint32 uiType         =(MIDF_TYPE_READ|MIDF_TYPE_WRITE);

    // Initialize the MIDF driver
    MIDF_FilterInitialize();

    // Register the memory region filter.
    MIDF_RegisterFilterRegion(uiMemoryBase, uiSize, uiID, uiType);

    // Unregister the memory region filter already registered
    MIDF_UnregisterFilterRegion(uiMemoryBase, uiSize, uiID, uiType);
}
```

7 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

8 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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